

Total No. of Questions : 6]

SEAT No. :

P78

[Total No. of Pages : 2

APR-17/BE/Insem.-90
B.E. (Computer Engineering)
CONCURRENCY ON OPEN SOURCE SYSTEMS
(2012 Pattern) (Semester - II) (Open Elective - IV)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Assume suitable data if necessary.*

- Q1)** a) What is concurrency? Explain concurrency as a modularization paradigm. [4]
b) Explain “Sleeping Barber” problem. [3]
c) Explain Android architecture with a block diagram. [3]

OR

- Q2)** a) Explain “Dining Philosopher’s” problem. [3]
b) Explain solution to “Dining Philosopher’s” problem using semaphores. [4]
c) What are the challenges in concurrency? [3]

- Q3)** a) Explain syntax structure of the CCS calculus. [4]
b) Explain the working of RMI. [3]
c) Write a note on execution of co-routines. [3]

OR

- Q4)** a) Explain various activation techniques of a procedure call. [4]
b) Explain deep calls with examples. [3]
c) Explain Android approach to program expression. [3]

P.T.O.

Q5) a) State similarities and differences between sequential and concurrent. Execution. [5]

b) Write an algorithm for converting a program expression to realize concurrent program execution model. [5]

OR

Q6) a) Explain context for program execution with an example. [5]

b) Explain the concurrent and distributed program execution model. [5]

